

Who Judges the Judge in RAG?

Controlled Pairing Effects in LLM Evaluation of Induced Factual Errors

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Abstract

LLM-as-a-judge evaluation is now common for retrieval-augmented generation (RAG), yet the judge is often drawn from the same model family as the system being evaluated. We introduce Eval-Pair Matrix, a controlled RAG meta-evaluation protocol: starting from GaRAGe questions and grounding passages, we induce hidden source-relative contradictions, ask GPT, Grok, and Gemini deployments to answer from the perturbed grounding, and then ask the same three deployments as blind judges to evaluate those answers against the original grounding. The full experiment contains 300 core records, 897 labeled generator outputs, and 2,683 judge verdicts in a crossed 3×3 matrix. To address a key validity concern, we treat the 275 fully validated records as the primary analysis set and replace raw diagonal-vs-off-diagonal comparisons with an answer-paired estimand: on the exact same candidate answer, does the matching judge behave differently from the two non-matching judges? The answer-paired result does not support broad self-leniency. Raw diagonal and off-diagonal F1 are similar (84.4 diagonal vs. 83.4 off-diagonal), and the paired same-deployment recall effect is near zero (-0.5 pp, 95% cluster-bootstrap CI $[-2.7, +1.7]$). The only robust paired difference is lower flagging by matching judges on answers that avoided the induced false claim (-4.3 pp, CI $[-6.6, -2.0]$). A targeted 88-case human evaluation finds no genuine false alarms among reviewed apparent false positives: they are real source errors caught by the judge or mistakes in the induced-error adoption labels. The resulting lesson is methodological: RAG judge studies should report full judge-by-generator matrices, but claims about self-leniency require answer-level pairing, behavior stratification, and careful alignment between induced-error adoption labels and judge tasks that flag any factual error.

1 Introduction

Retrieval-augmented generation (RAG) systems combine a language model’s parametric knowledge with external passages supplied at inference time (Lewis et al., 2020). This makes them attractive for knowledge-intensive applications, but it also creates a difficult evaluation problem: an answer may be fluent, cited, and still contradict the reference passages. In practice, many RAG evaluations now use another LLM as the judge because LLM judges are cheap, fast, and can emit structured rationales (Liu et al., 2023; Zheng et al., 2023; Kim et al., 2023). When the generator and judge are from the same model family, evaluation may become circular.

Existing LLM-as-a-judge work documents position, verbosity, length, style, familiarity, and self-preference effects (Dubois et al., 2024; Shi et al., 2024; Liu et al., 2024; Wataoka et al., 2024; Ye et al., 2024; Spiliopoulou et al., 2025). But open-ended preference settings make self-preference hard to isolate: a judge may prefer its own model’s output because it is better, clearer, more familiar, or simply more stylistically aligned. We study the question in a controlled factual setting. If we can induce a known source-relative contradiction hidden from the judge, then the judge is not choosing a favorite answer; it is detecting whether a candidate answer contradicts the original evidence.

We build the task from GaRAGe, a benchmark with human-written answers and passage-level grounding annotations (Sorodoc et al., 2025). For each selected example, an LLM perturber chooses one answer-causal target proposition and rewrites every passage that states or entails that claim so the evidence consistently supports a plausible but false replacement value. Generators answer the original question using the perturbed grounding. Judges then evaluate the answer against the unmodified grounding only. Figure 1 summarizes the end-to-

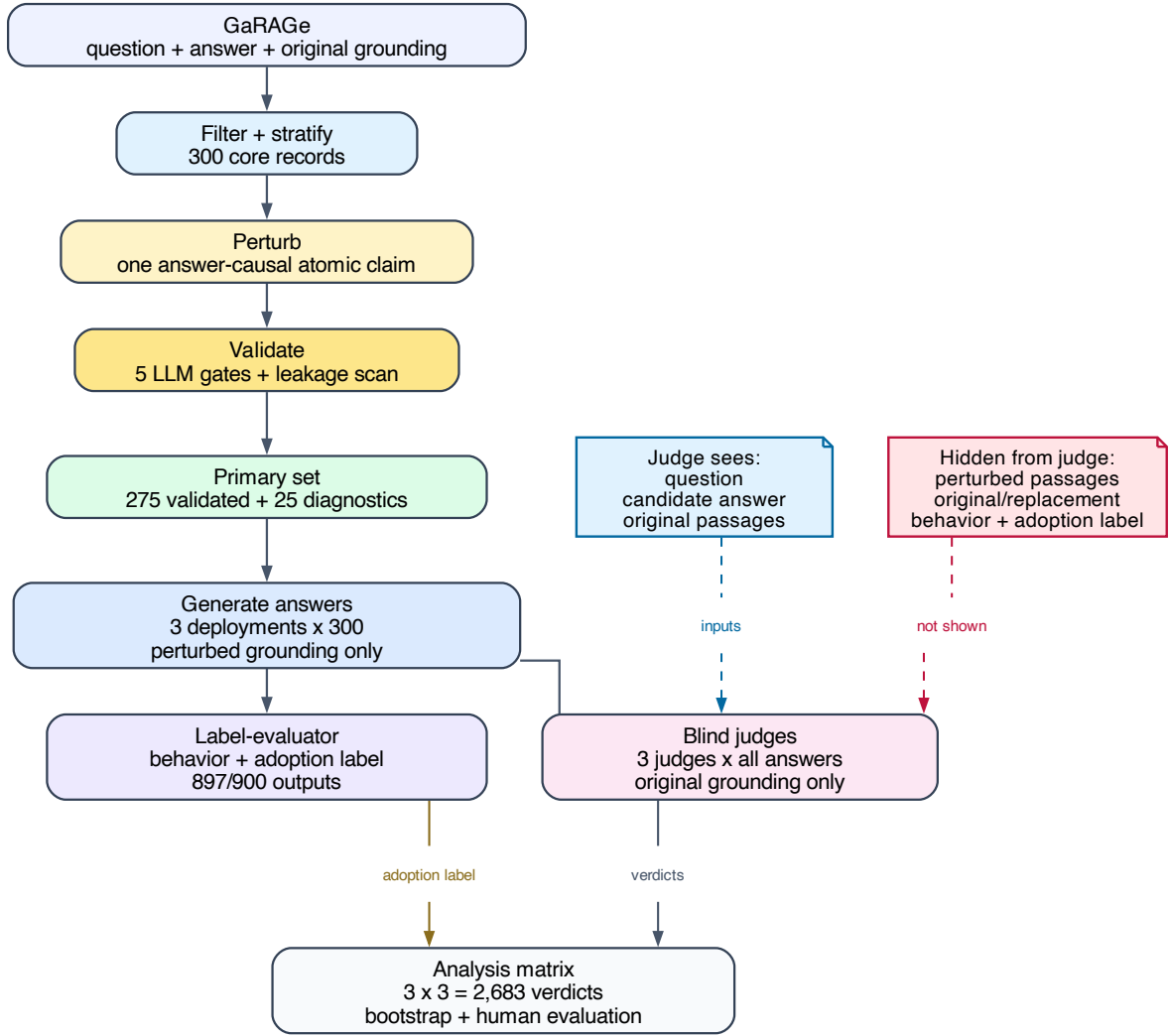


Figure 1: The Eval-Pair Matrix pipeline. Perturbations are generated and validated before answer generation. Judges see only the original question, candidate answer, and original grounding; they do not see the perturbation, behavior label, or induced-error adoption label. The final analysis combines the crossed matrix, answer-paired bootstrap, behavior stratification, and a targeted human evaluation.

end design.

A raw diagonal analysis compares each judge’s same-family cell with that judge’s two off-diagonal generator cells. That contrast is useful descriptively but not inferentially: it compares different answers with different styles, refusal rates, and difficulty. The paired estimand instead asks: *on the exact same candidate answer*, does the matching judge behave differently from the non-matching judges? This reframes the study from provider-specific answer distributions to same-deployment pairing effects.

This paired analysis changes the interpretation of the raw matrix. On the 275 fully validated records, the raw diagonal/off-diagonal F1 difference is small (84.4 diagonal vs. 83.4 off-diagonal). A paired answer-level cluster bootstrap finds no ro-

bust same-deployment effect on error-detection recall ($\Delta = -0.5$ pp, 95% CI $[-2.7, +1.7]$): judges are not broadly less likely to catch induced errors in answers generated by their own deployment. The only robust paired difference appears on answers that avoided the induced false claim: the judge from the same deployment as the generator flags these answers less often than the two other judges ($\Delta = -4.3$ pp, CI $[-6.6, -2.0]$). This should not be read as a lower false-alarm rate. Our automatic induced-error adoption label records only whether the answer adopted the false claim introduced by the perturbation, while the judge is asked to flag any factual error against the original passages. An answer can avoid the induced false claim and still contain another source-relative error, especially when it refuses, hedges, or describes inconsistent evi-

dence. In the targeted 88-case human evaluation, none of the 25 reviewed apparent false positives were genuine false alarms: they were alternate real source errors, mistakes in the induced-error adoption label, or unclear cases. Thus the apparent false-positive-rate effect is best interpreted as a mismatch between the adoption label and the judge’s broader source-grounding task, not as evidence of same-deployment self-leniency.

Contributions. The paper’s contributions are both methodological and empirical:

- **Protocol:** a controlled RAG meta-evaluation setup with hidden, localized source contradictions and blind judging against the original grounding.
- **Paired estimand:** an answer-level same-deployment contrast that compares matching and non-matching judges on the same candidate answer.
- **Empirical result:** no robust evidence that same-deployment judges broadly miss induced errors in their own deployment’s answers.
- **Interpretation:** the remaining flagging gap reflects generator behavior and label/task mismatch, supported by targeted human evaluation.

2 Related Work

RAG and grounding evaluation. RAG combines parametric generation with retrieved non-parametric evidence (Lewis et al., 2020). Automated RAG evaluators such as RAGAS and ARES score dimensions including faithfulness, context relevance, and answer correctness (Es et al., 2023; Saad-Falcon et al., 2023). RAGTruth and FaithBench provide hallucination annotations or challenging factuality cases (Niu et al., 2024; Bao et al., 2024). We use GaRAGe as the source dataset because it provides long-form answers and passage-level grounding over 2,366 questions and more than 35K passages (Sorodoc et al., 2025). Our protocol is narrower than general RAG scoring: it creates paired original/perturbed grounding sets so the target contradiction is localized and hidden from the judge.

Contextual judges and refusal behavior. LLM judges are widely used to evaluate generated text and instruction-following outputs, including G-Eval, MT-Bench, Chatbot Arena, and Prometheus (Liu et al., 2023; Zheng et al., 2023; Kim et al.,

2023). Recent contextual judge benchmarks show that evaluating responses against context remains difficult. ContextualJudgeBench evaluates contextual judges with 2,000 response pairs and explicitly includes RAG-QA faithfulness and refusal behavior (Xu et al., 2025). RAGferee trains RAG-centric contextual reward models for groundedness and appropriate refusals (Coman et al., 2025). These studies motivate our behavior analysis: apparent false positives concentrate in refusals and conflict-aware outputs, not in ordinary committed answers.

Self-preference and controlled judge meta-evaluation. Prior work documents model-judge biases, including length, position, style, familiarity, and self-preference effects (Dubois et al., 2024; Shi et al., 2024; Liu et al., 2024; Wataoka et al., 2024; Ye et al., 2024). A central difficulty is identification: a judge may favor its own model’s output because of true quality, stylistic familiarity, or family-specific conventions rather than self-leniency. Recent work therefore argues for bias estimates that control for response quality or independent judgments (Spiliopoulou et al., 2025; Chen et al., 2025a). In fact-centric RAG, Chen et al. (2025b) find no significant self-preference effect, suggesting that factual grounding may reduce the style-driven effects seen in open-ended preference tasks. Our work tests the same concern in a stricter source-grounded setting: we induce hidden contradictions in retrieved passages, generate natural answers from the perturbed grounding, and compare matching versus non-matching judges on the exact same candidate answer. This answer-paired design separates same-deployment judging effects from generator quality, refusal rates, and answer difficulty. REFLECT also uses controlled localized interventions for judge meta-evaluation, but studies evidence-based research-agent traces rather than RAG answer generation (Wang et al., 2026).

3 Task and Data Construction

3.1 Base Dataset and Perturbation Design

The pipeline starts from GaRAGe examples with a valid information-seeking question, a validated human-written answer, and at least two grounding passages labeled as answering the question. For each selected record, an LLM perturber receives the question, the human answer, and answer-bearing plus related grounding passages. The perturber

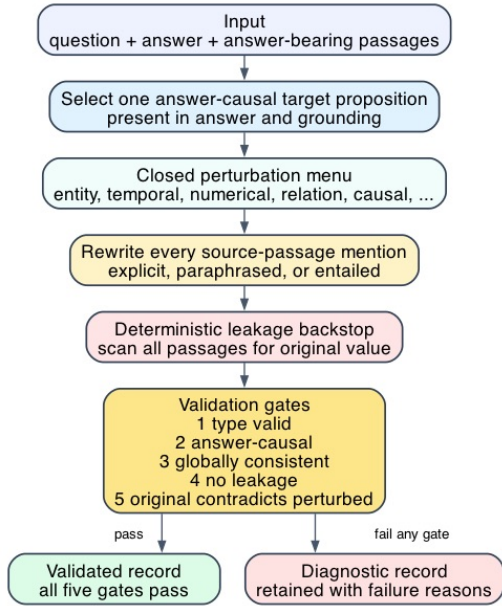


Figure 2: Perturbation and validation mechanics. A perturbation is valid for the primary analysis only if all five gates pass. Failed records are retained only as diagnostics.

must select exactly one answer-causal target proposition: a claim present in the answer and grounding such that changing it would change the answer to the question.

The closed perturbation menu includes entity substitutions, temporal shifts, numerical shifts, relation inversions, location and affiliation changes, ranking flips, causal changes, negation/modality changes, and multihop bridges. For every passage sent to the perturber, the output includes a full rewritten passage if the passage states or entails the selected claim, or a skip reason otherwise. The rewrite must remove the original value and any anchors that leak it. The selected 300-record pool is intentionally mixed rather than type-balanced; Figure 3 reports the type distribution, validation split, and perturber-specific type mix.

Perturbation examples. A numerical-shift record asks: “How did Caitlin Clark’s WNBA debut influence league attendance in 2024?” The original answer-causal claim is that the debut led to a 48% increase in overall league attendance. The perturber changes the value to 35%, producing a passage that says the league saw a 35% jump while preserving the rest of the passage. A faithful generator reading only the perturbed passage should answer with 35%, but that answer is false relative to the original passage.

A relation-inversion record is more semantically disruptive. The original Zantac-trial claim says a jury returned a defense verdict, producing a win and rising stock prices. The perturbed claim says the jury returned a plaintiff verdict, producing a loss and plummeting stock prices. This illustrates why relation inversions require consistency edits around the target proposition rather than a single-token replacement.

3.2 Validation and Diagnostic Records

Each perturbation is validated by an LLM validation pass plus deterministic leakage checks. The five gates are type validity, answer causality, global context consistency, no original-answer leakage, and original contradicts perturbed. The final evaluation pool has 300 unique core records selected from perturbations proposed by GPT, Grok, and Gemini perturbers: 83, 92, and 125 selected records, respectively. Of these, 275 pass all five gates and 25 are failed-validation diagnostic records. We use the 275 validated records for every main result and report the 25 diagnostics only as a sensitivity analysis. Figure 3 shows that diagnostics are concentrated by count in entity and numerical records, with larger diagnostic shares in the smaller location and ranking strata.

4 Generation, Labels, and Judging

4.1 RAG Generation from Perturbed Grounding

Each generator receives the original question and the perturbed grounding passages. The prompt instructs the model to answer using only the provided passages, cite passage IDs supporting factual claims, and refuse if the evidence is insufficient or inconsistent. We run three generator deployments over all 300 records, yielding 900 intended generator outputs and 897 outputs with LLM behavior labels.

We use six generator-behavior labels:

Context follow: asserts the perturbed value.

Memory override: asserts the original value.

Both claims: includes both original and perturbed values.

Conflict awareness: reports inconsistent evidence.

Refusal/insuff.: refuses or says the evidence is insufficient.

Unrelated/failed: gives an off-task or unusable answer.

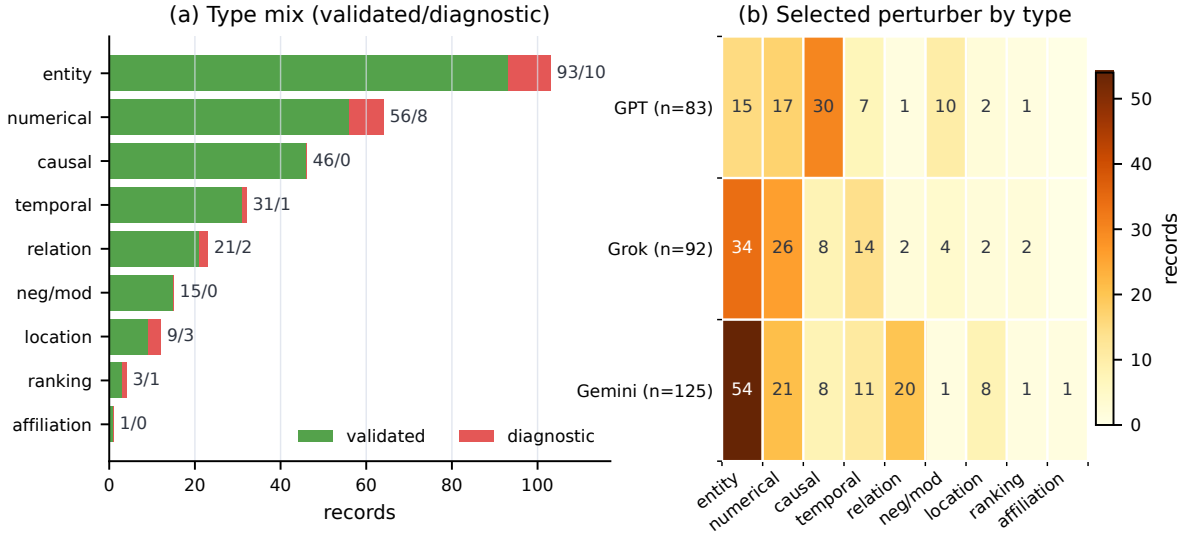


Figure 3: Composition of the 300 selected perturbation records. Panel (a) shows perturbation-type counts, with each row labeled as validated/diagnostic. Panel (b) shows the selected perturber-by-type matrix; row totals give the number of selected records proposed by each perturber.

Behavior label	GPT	Grok	Gemini
Context follow	153 (51.2%)	141 (47.2%)	143 (47.8%)
Memory override	10 (3.3%)	8 (2.7%)	4 (1.3%)
Both claims	20 (6.7%)	15 (5.0%)	18 (6.0%)
Conflict aware	3 (1.0%)	12 (4.0%)	4 (1.3%)
Refusal/insuff.	16 (5.4%)	74 (24.7%)	54 (18.1%)
Unrelated/failed	97 (32.4%)	49 (16.4%)	76 (25.4%)

Table 1: Generator behavior labels over 897/900 LLM-labeled outputs. Percentages use the labeled denominator for each generator (299).

A separate LLM label-evaluation step assigns these labels using paraphrase-aware criteria; deterministic string matching is retained only as a diagnostic signal.

The share of non-context-following outputs differs by model. Grok refuses or declares insufficiency far more often than GPT, while GPT more often produces an answer that avoids the specific perturbed span. These behavior differences later explain much of the judge matrix.

4.2 Two-Layer Labeling and Blind Judging

The experiment has two LLM-evaluation layers. Layer 1 is a single GPT-5.4 label-evaluator that sees the answer and perturbation metadata and returns a behavior label plus `entails_perturbed_claim`. The induced-error adoption label is derived from that entailment flag: did the answer adopt the false value introduced by the perturbation? Layer 2 consists of the three crossed matrix judges. They see only the question, candidate answer, and original passages. They are blind to the perturbed ground-

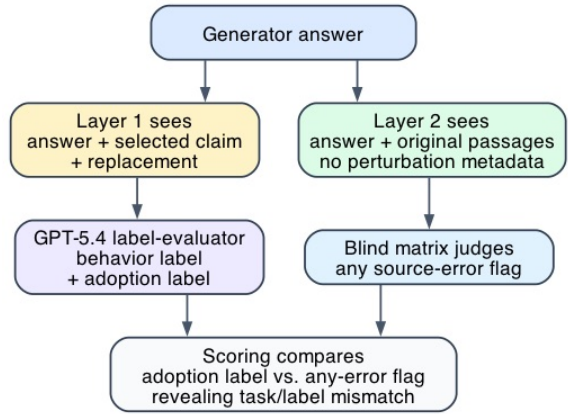


Figure 4: Two-layer setup. The induced-error adoption label comes from a GPT-5.4 label-evaluator with perturbation access; the matrix judges are blind and are asked for any source-relative factual error. This task/label mismatch is central to interpreting apparent false positives.

ing, original value, replacement value, generator behavior label, and induced-error adoption label.

For judging, each judge outputs a verdict, a boolean factual-error flag, the wrong claim if any, the most directly contradicting source passage ID, an explanation, and confidence. The main run uses one exact deployment per provider family: GPT-5.4, Grok 4.3, and Gemini 3.5 Flash. Because there is no same-provider/different-deployment condition, we use *same-deployment pairing effect* rather than same-model-family bias.

Judge	GPT answers			Grok answers			Gemini answers			Row F1
	P	R	F1	P	R	F1	P	R	F1	
GPT	94.0	78.1	85.3	77.9	81.6	79.7	81.6	83.2	82.4	82.5
Grok	90.0	77.8	83.4	80.1	79.6	79.9	85.8	85.8	85.8	83.0
Gemini	89.2	86.4	87.8	78.1	85.0	81.4	85.4	90.6	87.9	85.7
Col F1	85.5			80.3			85.4			

Table 2: Validated-only matrix (275 validated records; diagonal = same-deployment cells, bold). Per cell: precision/recall/F1 (%). Mean diagonal F1 = 84.4, mean off-diagonal F1 = 83.4.

5 Results

5.1 Validated Judge Matrix

Table 2 shows the validated-only judge-by-generator matrix. Across cells, F1 ranges from 79.7% to 87.9%. The diagonal is not obviously inflated: mean diagonal F1 is 84.4 and mean off-diagonal F1 is 83.4. Grok-generated answers are the hardest column for every judge, showing that raw diagonal-vs-off-diagonal comparisons mix possible same-deployment effects with generator-column difficulty.

5.2 Answer-Paired Same-Deployment Effects

Every candidate answer is judged by all three judges, so we can compare judges on the same answer rather than comparing different generator columns. For answer a generated by deployment g , the matching judge is g and the two non-matching judges are the other deployments. Let $Y_{j,a}$ be judge j 's binary flag. The same-answer contrast is

$$\Delta_a = Y_{g,a} - \frac{1}{2} \sum_{j \neq g} Y_{j,a}.$$

A positive Δ_a means the matching judge flags more often than the other two judges on that answer; a negative value means it flags less often. We average this contrast three ways: over answers that adopted the induced false claim (recall Δ), over answers that avoided that claim (avoided-claim flag Δ), and over all answers (all-answer flag Δ). Confidence intervals use a cluster bootstrap by core record ID with 10,000 replicates.

Table 3 gives the paired estimates. The pre-specified global recall effect is near zero (-0.5 pp, 95% CI $[-2.7, +1.7]$), so the paired analysis does not support the hypothesis that same-deployment judges broadly miss induced errors in their own deployment's answers. The robust negative avoided-claim flag effect means that matching judges flag answers that avoided the induced false claim less often than the two non-matching judges. This should

not be read as a lower false-alarm rate, because these answers can still contain other source-relative errors; Section 5.4 audits that mismatch. Figure 6 in the appendix shows how the raw diagonal signature changes when the comparison is paired.

Set	Recall Δ	Avoided-claim flag Δ	All-answer flag Δ
<i>Confirmatory</i>			
Global	-0.5 [-2.7,+1.7]	-4.3 [-6.6,-2.0]	-2.2 [-3.8,-0.6]
<i>Exploratory, by generator</i>			
GPT	-3.8 [-8.0,+0.3]	-6.7 [-11.4,-2.6]	-5.0 [-8.1,-2.0]
Grok	-3.7 [-8.7,+0.7]	-4.7 [-9.3,-0.4]	-4.2 [-7.5,-1.1]
Gemini	+6.0 [+2.9,+9.6]	-1.6 [-4.7,+1.6]	+2.6 [+0.2,+5.0]

Table 3: Paired same-deployment effect: matching judge minus the mean of the two non-matching judges on the same answer. Each numeric cell is Δ [95% cluster-bootstrap CI] in percentage points. Per-generator rows are exploratory and uncorrected.

5.3 Generator Behavior Explains the Matrix

Table 4 stratifies judge outcomes by generator behavior. Here TP/FP/FN/TN are computed against the induced-error adoption label: TP = adopted and flagged, FP = avoided but flagged, FN = adopted but not flagged, and TN = avoided and not flagged. Recall is concentrated in context-following answers: 1,025 of 1,139 true positives come from context-follow outputs. Apparent false positives are concentrated in conflict-aware and refusal/insufficient answers: 176 of 209 come from those two behavior classes. Memory override provides a useful control: when the generator uses the original value, judges almost never flag an error.

5.4 Human Evaluation of Apparent False Positives

The avoided-claim flagging gap in Table 3 creates an interpretive question: when the induced-error adoption label calls a judge flag a false positive, is the judge actually wrong? In our setup, an apparent false positive is an answer that avoids the induced false claim but is still flagged by the judge. That label is narrower than the judge task, because judges

Generator behavior	verdicts	answers	TP	FP	FN	TN	Headline rate [95% CI]
context-follow	1212	405	1025	0	187	0	recall 84.6 [81.5,87.6]
both-claims	135	45	88	0	44	3	recall 66.7 [54.5,78.8]
memory-override	42	14	0	0	0	42	FPR 0.0 [0.0,7.1]
conflict-aware	54	18	3	46	0	5	FPR 90.2 [76.5,100.0]
refusal/insuff.	400	134	23	130	1	246	FPR 34.6 [27.4,42.2]
unrelated/failed	606	202	0	25	0	581	FPR 4.1 [2.1,6.3]
unlabeled	9	3	0	8	0	1	FPR 88.9 [66.7,100.0]

Table 4: Judge outcomes pooled over all nine cells, stratified by LLM behavior labels. The reported rate is recall for answers that adopted the induced false claim and apparent false-positive rate otherwise.

are asked to flag *any* factual error against the original passages. An answer can avoid the induced false claim and still contain another source-relative error.

We therefore reviewed a targeted behavior-by-cell queue, capped at 12 cases per populated stratum, and adjudicated 88 of 153 sampled cases. The key result is that none of the 25 reviewed apparent false positives were genuine false alarms: 22 were valid detections of alternate source errors, 2 were mistakes in the induced-error adoption label, and 1 was unclear.

Automatic case	Reviewed	Human finding	Interpretation
Apparent FP	25	0 genuine false alarms	mostly alternate source errors or adoption-label mistakes
TN	31	31 genuine TN	no missed errors found in this control set
TP	23	21 genuine TP	recall labels mostly valid
FN	9	6 genuine misses	some adoption labels too broad

Table 5: Human evaluation summary. Rule-of-three upper bounds are about 12% for genuine false alarms among reviewed apparent FPs and about 10% for missed errors among reviewed TNs.

This supports a different interpretation from self-leniency. The robust avoided-claim gap is better read as a label/task mismatch: the adoption label asks whether the answer took up the induced false claim, while the judge is allowed to catch any source-relative error. The SEC/FASB case below shows the distinction. The answer avoids the induced SEC claim, but it makes a separate source-grounding error by attributing an ASC 842 lease-accounting effect to recent SEC changes.

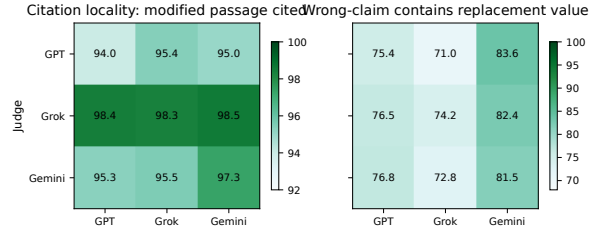


Figure 5: Localization metrics from the raw judge verdicts. Passage-level localization is consistently high; exact replacement containment in the wrong-claim text is lower because judges often paraphrase the erroneous claim.

Worked apparent-FP example: Grok judge on GPT answer.	
Question	How have recent SEC changes affected operating lease reporting practices?
Induced claim	The perturbed SEC passage replaced “proactive initiatives targeting compliance failures” with “detailed audits of financial reporting accuracy.”
GPT answer	It did not adopt that induced SEC claim. Instead, it said recent changes made operating lease reporting more balance-sheet focused and cited ASC 842 lease-accounting requirements.
Grok flag	Valid under the judge task: the source attributes ASC 842 to FASB, and no source links recent SEC changes to those balance-sheet lease requirements.
Why FP?	The adoption label says “avoided induced claim,” while the judge says “contains another source-relative error.” The automatic matrix counts this as an apparent FP, but human evaluation treats it as an alternate valid error detection.

Thus, under the any-error judge task, this is a valid detection; under the narrower induced-error adoption label, it is counted as an apparent FP.

5.5 Localization Remains Strong

For downstream debugging, a judge is more useful when it points to the contradicted source. Figure 5 summarizes full-set localization metrics computed from the full judge output set. When judges flag an error, their supporting passage lies in the known modified passage set 94.0–98.5% of the time. Wrong-claim strings contain the replacement value 71.0–83.6% of the time, reflecting paraphrase rather than passage-localization failure.

6 Discussion

Raw matrices are valuable but not sufficient. The full judge-by-generator matrix reveals hetero-

geneous generator difficulty and judge behavior. However, raw diagonal-vs-off-diagonal contrasts compare different answers. The paired estimand changes the scientific claim: after controlling for the answer by comparing judges on the same candidate response, the global recall effect is near zero.

The apparent FPR effect is a label/task mismatch. The robust avoided-claim flagging gap is not evidence that judges over- or under-flag errors on same-deployment outputs. It is primarily a mismatch between an induced-error adoption label and an any-error judge task. The human evaluation found no genuine false alarms among 25 reviewed apparent FPs. This matters for future controlled benchmarks: if the labels record only whether an answer adopted the induced false claim, then unrestricted judge flags should be adjudicated into induced-error detection, alternate valid error detection, and genuine false alarm.

Behavior is a central axis of RAG judge evaluation. Generator behavior explains much of the matrix. Context-following answers create ordinary induced-error recall cases. Refusals and conflict-aware answers create apparent false positives because they avoid the induced claim while discussing source insufficiency or inconsistency in ways that may contradict the original grounding. Benchmarks that pool committed answers with refusals and hedges can obscure the actual failure mode.

Implications for practice. LLM judges can be useful source-grounding evaluators: the matrix shows strong precision/recall and high passage localization. But evaluations that reuse a model family as both generator and judge should report the full matrix, behavior strata, and answer-paired same-vs-cross contrasts. Aggregated same-model metrics are not enough, especially when refusal behavior differs by generator.

7 Conclusion

We introduced Eval-Pair Matrix, a controlled RAG meta-evaluation protocol that induces hidden source contradictions and evaluates a full 3×3 judge-by-generator matrix. Together, the matrix, paired analysis, behavior stratification, and human evaluation support a narrower conclusion. On 275 validated records, there is no robust same-deployment recall effect after answer-level pairing. The only robust difference is lower flagging

by matching judges on answers that avoided the induced false claim, but a targeted human evaluation finds no genuine false alarms among reviewed apparent false positives. The broader lesson is methodological: in RAG judge studies, full matrices are useful, but claims about self-leniency require answer-level pairing, behavior stratification, and careful alignment between induced-error adoption labels and the judge task.

Limitations

The dataset is controlled and moderate in scale: 275 validated records and 25 failed-validation diagnostics. We evaluate one exact deployment per provider, so the design cannot separate exact-deployment effects from provider-family or style-familiarity effects. The behavior labels and induced-error adoption labels come from a single GPT-5.4 label-evaluator rather than independent human annotation; the human evaluation found low but nonzero internal contradiction and about 8% adoption-label issues in reviewed refusal/conflict cases. The human evaluation is targeted and single-adjudicator, not a double-annotated study with agreement statistics. Judge calls are single-shot, so stochastic flip rates are not estimated. Finally, the strict source-grounding stance used in the human evaluation is defensible but consequential: an alternative adoption-only interpretation would classify some conflict/refusal flags as artifacts rather than alternate errors, though not as genuine false alarms.

Ethics Statement

The work creates controlled factual perturbations for evaluation. Perturbed passages should not be presented as factual information, and any release should preserve provenance, validation metadata, and original/perturbed labels. The benchmark does not evaluate people or make consequential decisions about individuals. The main risk is misuse of the perturbation pipeline to produce misleading text; the mitigation is to release it only as a traceable evaluation artifact with clear documentation that the corrupted passages are synthetic counterfactuals.

References

Forrest Sheng Bao, Miaoran Li, Renyi Qu, Ge Luo, Erana Wan, Yujia Tang, Weisi Fan, Manveer Singh Tamber, Suleman Kazi, Vivek Sourabh, Mike Qi, Ruixuan

- Tu, Chenyu Xu, Matthew Gonzales, Ofer Mendelevitch, and Amin Ahmad. 2024. FaithBench: A diverse hallucination benchmark for summarization by modern LLMs. *arXiv preprint arXiv:2410.13210*.
- Zhi-Yuan Chen, Hao Wang, Xinyu Zhang, Enrui Hu, and Yankai Lin. 2025. Beyond the Surface: Measuring Self-Preference in LLM Judgments. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 1653–1672, Suzhou, China. Association for Computational Linguistics. DOI: 10.18653/v1/2025.emnlp-main.86.
- Yen-Shan Chen, Jing Jin, Peng-Ting Kuo, Chao-Wei Huang, and Yun-Nung Chen. 2025. LLMs are Biased Evaluators But Not Biased for Fact-Centric Retrieval Augmented Generation. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 26669–26684, Vienna, Austria. Association for Computational Linguistics. DOI: 10.18653/v1/2025.findings-acl.1369.
- Andrei C. Coman, Ionut-Teodor Sorodoc, Leonardo F. R. Ribeiro, Bill Byrne, James Henderson, and Adrià de Gispert. 2025. RAGferee: Building Contextual Reward Models for Retrieval-Augmented Generation. *arXiv preprint arXiv:2509.26011*.
- Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. 2024. Length-controlled AlpacaEval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2023. RAGAS: Automated evaluation of retrieval augmented generation. *arXiv preprint arXiv:2309.15217*.
- Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and Minjoon Seo. 2023. Prometheus: Inducing fine-grained evaluation capability in language models. *arXiv preprint arXiv:2310.08491*.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*.
- Yang Liu, Dan Iter, Yichong Xu, Shuhang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-Eval: NLG evaluation using GPT-4 with better human alignment. *arXiv preprint arXiv:2303.16634*.
- Yiqi Liu, Nafise Sadat Moosavi, and Chenghua Lin. 2024. LLMs as narcissistic evaluators: When ego inflates evaluation scores. In *Findings of the Association for Computational Linguistics: ACL 2024*.
- Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, Kashun Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. *arXiv preprint arXiv:2401.00396*.
- Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2023. ARES: An automated evaluation framework for retrieval-augmented generation systems. *arXiv preprint arXiv:2311.09476*.
- Lin Shi, Chiyu Ma, Wenhua Liang, Weicheng Ma, and Soroush Vosoughi. 2024. Judging the judges: A systematic study of position bias in LLM-as-a-judge. *arXiv preprint arXiv:2406.07791*.
- Ionut Teodor Sorodoc, Leonardo F. R. Ribeiro, Rexhina Blloshmi, Christopher Davis, and Adrià de Gispert. 2025. GaRAGe: A Benchmark with Grounding Annotations for RAG Evaluation. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 17030–17049, Vienna, Austria. Association for Computational Linguistics. DOI: 10.18653/v1/2025.findings-acl.875.
- Evangelia Spiliopoulou, Riccardo Fogliato, Hanna Burnsky, Tamer Soliman, Jie Ma, Graham Horwood, and Miguel Ballesteros. 2025. Play favorites: A statistical method to measure self-bias in LLM-as-a-judge. *arXiv preprint arXiv:2508.06709*.
- Leyao Wang, Yanan He, Peng Chen, Asaf Yehudai, Yixin Liu, Rex Ying, Michal Shmueli-Scheuer, and Arman Cohan. 2026. Time to REFLECT: Can We Trust LLM Judges for Evidence-based Research Agents? *arXiv preprint arXiv:2605.19196*.
- Koki Wataoka, Tsubasa Takahashi, and Ryokan Ri. 2024. Self-preference bias in LLM-as-a-judge. *arXiv preprint arXiv:2410.21819*.
- Austin Xu, Srijan Bansal, Yifei Ming, Semih Yavuz, and Shafiq Joty. 2025. Does Context Matter? ContextualJudgeBench for Evaluating LLM-based Judges in Contextual Settings. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9541–9564, Vienna, Austria. Association for Computational Linguistics. DOI: 10.18653/v1/2025.acl-long.470.
- Jiayi Ye, Yanbo Wang, Yue Huang, Dongping Chen, Qihui Zhang, Nuno Moniz, Tian Gao, Werner Geyer, Chao Huang, Pin-Yu Chen, Nitesh V. Chawla, and Xiangliang Zhang. 2024. Justice or prejudice? Quantifying biases in LLM-as-a-judge. *arXiv preprint arXiv:2410.02736*.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *arXiv preprint arXiv:2306.05685*.

Example	Type	Question	Value swap	Perturbed passage snippet	Validation gates
numeric	numerical shift	How did Caitlin Clark’s WNBA debut influence league attendance in 2024?	48% → 35%	In her rookie WNBA season, the league saw a 35% jump in attendance from 2023. The league’s total attendance of 2,353,735 was the highest in 22 seasons.	all five pass
relation	relation inversion	What were the outcomes of the initial Zantac trials involving Abbott and Mead Johnson?	defense verdict → plaintiff verdict	The plaintiff verdict in Whitfield v. Abbott is a devastating loss for Abbott and Reckitt, Mead Johnson’s parent—whose share prices plummeted significantly after the news.	all five pass
diagnostic	numerical shift	What are the expected benefits and risks of Ethereum 2.0 for long-term investors?	two → seventeen	Ethereum’s upcoming Pectra upgrade in 2025 is met with optimism, as seventeen out of its 19 upgrades historically generated notable price impacts.	fails type validity, answer causality, and leakage

Table 6: Representative perturbation and validation cases. The diagnostic example shows why the 25 failed-validation records are not used for the main conclusions.

Example	Behavior	Question	Swap	Generator answer snippet	Label rationale
context follow	context follow	How did Caitlin Clark’s WNBA debut influence league attendance in 2024?	48% → 35%	Caitlin Clark’s 2024 WNBA debut/arrival was associated with a major jump in league attendance. Leaguewide attendance rose 35% from 2023, reaching 2,353,735 total fans.	The answer explicitly states 35%, matching the perturbed passages and not the original 48%.
memory override	memory override	How does Loschmidt’s paradox challenge Boltzmann’s H theorem?	reversible → irreversible	Loschmidt’s paradox challenges Boltzmann’s H-theorem by arguing that one should not derive irreversible entropy increase from time-symmetric or microscopically reversible dynamics.	The answer states the standard formulation, contradicting the perturbed documents’ altered claim.
refusal	refusal/insufficient	How has Letsile Tebogo’s cultural heritage influenced his athletic career?	Botswana → Ghana	The provided passages do not contain sufficient information on how Letsile Tebogo’s cultural heritage has influenced his athletic career.	The answer declines to assert an influence because the passages are insufficient.
conflict aware	conflict awareness	What impact has National Dog Month had on dog adoption rates?	promotes → discourages adoption	The passages do not provide a clear adoption-rate impact. They say National Dog Month promotes adoption while other passages inconsistently claim it discourages adoption.	The answer explicitly reports inconsistent evidence.

Table 7: Representative generation outputs and behavior-label rationales. These examples show why paraphrase-aware LLM labels were used instead of strict string matching.

A Perturbation and Validation Examples

B Generation Examples

C Judge Schema and Examples

The judge output schema is:

```
{
  "verdict": "correct | incorrect | unclear",
  "contains_factual_error": true | false,
  "wrong_claim": "string or null",
  "supporting_source_passage_id": "int or null",
  "explanation": "one or two sentences",
  "confidence": "high | medium | low"
}
```

D Raw Diagonal Signatures Versus Paired Contrasts

The raw diagonal contrast compares a judge’s diagonal cell against that judge’s two cross-generator

cells. It is retained only as a descriptive diagnostic because it compares different answers. The paired contrast compares the matching judge with non-matching judges on the same answer.

E Full-Set Judge Metrics

F Localization Table

G Validated, Full, and Diagnostic Sensitivity

H Additional Data Visualizations

I Human Evaluation Codebook

The targeted human evaluation records two judgments per case. The first checks whether the automatic induced-error adoption label is correct: for

Example	Cell	Adoption label	Pred.	Question	Wrong claim	Judge explanation
TP same	Gemini/Gemini	True	True	How did Caitlin Clark's WNBA debut influence league attendance in 2024?	Overall league attendance jumped by 35% compared to 2023.	The answer claims 35%, while the original source states 48%.
FN	GPT/GPT	True	False	How does RL contribute to personalized learning experiences in education?	–	The judge says the answer is supported by the passages, missing the induced replacement.
Apparent FP	Grok/GPT	False	True	How have recent SEC changes affected operating lease reporting practices?	Under ASC 842, companies must recognize most leases on the balance sheet.	Passage 2 attributes ASC 842 to FASB rather than SEC, and no source links SEC rule changes to the specific balance-sheet recognition requirement.
TN	GPT/Grok	False	False	How has Letsile Tebogo's cultural heritage influenced his athletic career?	–	The passages mention Tebogo and Botswana's sports culture but do not explain how his cultural heritage influenced his career.

Table 8: Representative judge outcomes. The apparent FP is a key example: the answer avoids the induced false claim but appears to be a valid alternate source-error detection.

Judge	Gen.	<i>n</i>	Adopted	TP	FP	FN	TN	Prec.	Rec.	F1
GPT	GPT	297	173	134	9	39	115	93.7	77.5	84.8
GPT	Grok	299	159	131	38	28	102	77.5	82.4	79.9
GPT	Gemini	296	165	140	30	25	101	82.4	84.8	83.6
Grok	GPT	299	175	136	15	39	109	90.1	77.7	83.4
Grok	Grok	300	159	128	31	31	110	80.5	80.5	80.5
Grok	Gemini	296	164	142	22	22	110	86.6	86.6	86.6
Gemini	GPT	299	175	151	19	24	105	88.8	86.3	87.5
Gemini	Grok	300	159	136	37	23	104	78.6	85.5	81.9
Gemini	Gemini	297	165	151	25	14	107	85.8	91.5	88.6

Table 9: Full 300-record confusion counts and metrics from the raw judge run. Metrics are percentages. The main paper uses the 275 validated-record matrix.

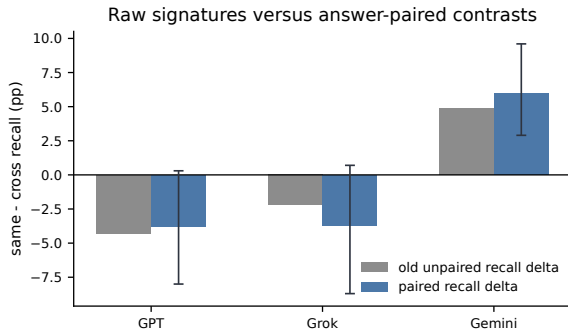


Figure 6: Raw diagonal recall deltas versus answer-paired recall deltas on the validated set. The paired analysis preserves some exploratory provider-level patterns but removes the basis for a broad self-lenient claim.

answers labeled as adopting the induced false claim, the evaluator checks whether the answer adopts that claim; for answers labeled as avoiding it, the evaluator checks whether the answer does avoid it. Mentioning a false value is not the same as adopting it; this distinction is critical for conflict-aware and refusal answers. The second judgment asks whether the judge’s flag/no-flag decision is

correct against the original passages under a strict source-grounding stance.

J Evaluation Implementation and Reproducibility

The paired analysis uses `paper/audit/paired_audit.py` with bootstrap seed 20260627 and 10,000 cluster replicates by core ID. The behavior table uses `paper/audit/behavior_stratified.py`, behavior labels from `llm_eval` only, and a rule-of-three upper bound for zero-event rows. The human-evaluation queue is built by `build_cell_review_queue.py` with seed 20260627 and a cap of 12 cases per populated behavior-by-cell stratum. The review application writes progress to disk, and `gen_audit_dossier.py` regenerates the summary tables and per-case log.

The primary run artifacts are `data/exp/3provider_300.jsonl`; three generation/label-evaluation runs under `data/runs/20260531-0051*`; and nine judge runs under `data/runs/20260531-*judge*`. Reproducibility checks verify 300 core records, 275 validated records, 897 labeled generator outputs,

Full-set confusion counts by judge x generator cell

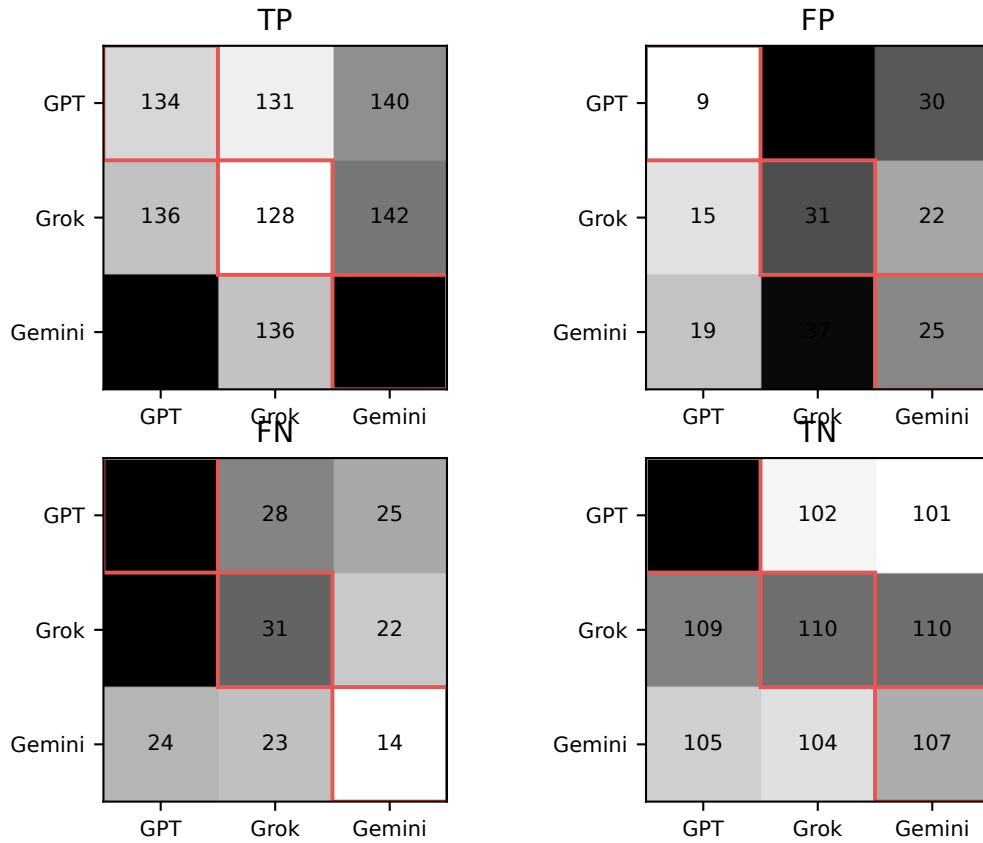


Figure 7: Full-set confusion counts per judge-generator cell. Diagonal cells are outlined in red.

Judge	Citation locality			Wrong claim contains replacement		
	Gen=GPT	Gen=Grok	Gen=Gemini	Gen=GPT	Gen=Grok	Gen=Gemini
GPT	94.0	95.4	95.0	75.4	71.0	83.6
Grok	98.4	98.3	98.5	76.5	74.2	82.4
Gemini	95.3	95.5	97.3	76.8	72.8	81.5

Table 10: Localization quality across matrix cells in the full-set analysis. Citation locality is consistently higher than exact replacement containment.

2,683 judge verdicts, and 894 same-deployment versus 1,789 cross-provider verdicts.

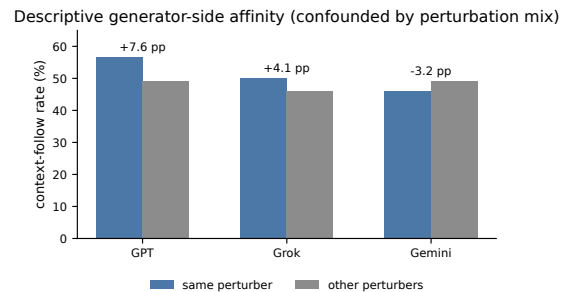


Figure 8: Generator-side self-affinity, retained as descriptive only. Same-perturber context-follow rates are confounded by provider-specific perturbation type preferences, summarized in Figure 3.

Set	Recall Δ	Avoided-claim flag Δ	Flag-rate Δ
Validated (275)	-0.5 [-2.7,+1.7]	-4.3 [-6.6,-2.0]	-2.2 [-3.8,-0.6]
Full (300)	-0.7 [-2.8,+1.4]	-4.2 [-6.3,-2.1]	-2.2 [-3.8,-0.7]
Diagnostic (25)	-2.4 [-8.8,+2.6]	-2.9 [-6.9,+0.0]	-2.7 [-6.7,+0.7]

Table 11: Global paired effects by dataset slice. The validated and full results agree; diagnostic records are small and uncertain.

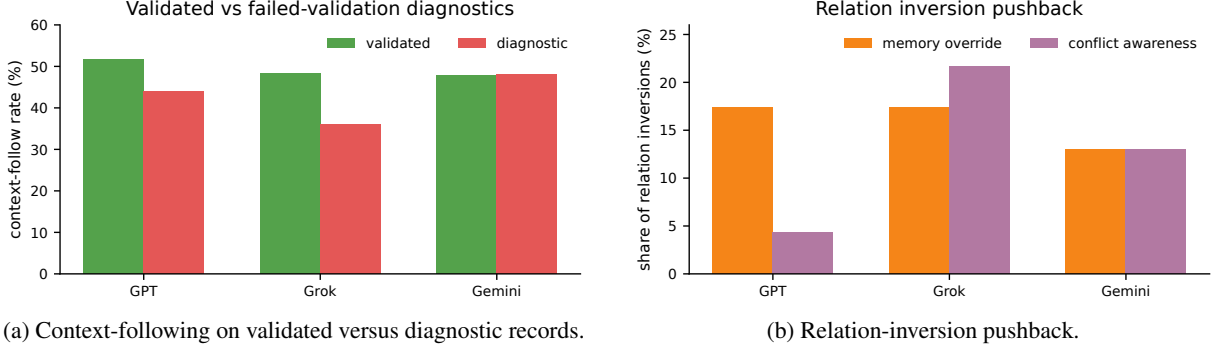


Figure 9: Additional generator behavior diagnostics. Diagnostic records still affect generation, while relation inversions reveal more memory override and conflict awareness than simpler substitutions.

Cell and evaluation calls	Derived outcome	Interpretation
TP, label ok, judge ok	genuine TP	judge caught a valid source error in an answer that adopted the induced false claim
TP, label ok, judge not ok FP, label ok, judge ok	right verdict, wrong reason alternate error	flag is correct but rationale/localization is not answer avoids the induced false claim but contains another valid source error
FP, label ok, judge not ok FP, label not ok	genuine false alarm adoption-label mistake	judge flagged without a source error answer actually adopted the induced false claim
FN, label ok, judge not ok FN, label not ok	genuine miss adoption label too broad	induced error was missed answer did not actually adopt the induced false claim
TN, label ok, judge ok	genuine TN	no induced error and no other source error found
TN, judge not ok	missed error	judge stayed silent despite a source error

Table 12: Human evaluation derivation rules. Apparent FPs in the automatic matrix can be false alarms, alternate valid source-error detections, or induced-error adoption-label mistakes.

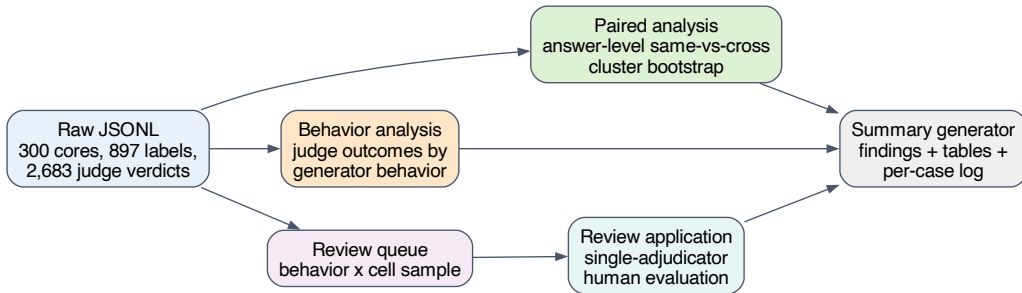


Figure 10: Evaluation implementation flow. Scripts are read-only over raw JSONL data and write only analysis artifacts, tables, and human-evaluation summaries.